



Before the Graph: The Making of Networks in Humanities Research

Network Analysis Workshop
University of Tartu – 6 May 2026

Dr. Matteo Romanello (UZH / ETHZ)

Where do edges come from?

Most of the work in network analysis happens before the algorithm runs.

	C	E	F	G	H
	is_annotated	text	person	location	at Matteo
1					
2		Le pape Jean VIII le reçut à sa communion, le reconnut, lui écrivit ; et malgré ce huitième concile oecuménique, qui avait	Judas	Rome	FALSE
3		anathématisé ce patriarche, le pape envoya ses légats à un autre concile à Constantinople, dans lequel Photius fut reconnu innocent	Judas	Constantinople	FALSE
4		par quatre - cent évêques, dont trois - cent l'avaient auparavant condamné. Les légats de ce même siège de Rome, qui l'avaient	patriarche Photius	Constantinople	TRUE
5		anathématisé, servirent eux - mêmes à casser le huitième concile oecuménique. Combien tout change chez les hommes ! Combien ce	patriarche Photius	Rome	FALSE
6		qui était faux, devient vrai selon les tems ! Les légats de Jean VIII s'écrivent en plein concile ; si quelqu'un ne reconnaît pas Photius,	patriarche Jean	Constantinople	FALSE
7		que son partage soit avec Judas. le concile s'écrit, longues années au patriarche Photius, et au patriarche Jean. enfin à la suite des	patriarche Jean	Rome	FALSE
8		actes du concile on voit une lettre du pape à ce savant patriarche, dans laquelle il lui dit ; nous pensons comme vous... etc. Il est donc	Michel Paléologue	Rome	FALSE
9		clair que l'église romaine et la grecque pensaient alors différemment de ce qu'on pense aujourd'hui. Il arriva depuis que Rome	Michel Paléologue	Constantinople	PROBABLE
10		adopta la procession du père et du fils ; et il arriva même qu'en 1274 l'empereur des grecs, Michel Paléologue, implorant contre les	Eugène IV	Rome	FALSE
11		turcs une nouvelle croisade, envoya au second concile de Lyon, son patriarche et son chancelier, qui chantèrent avec le concile en	Eugène IV	Constantinople	FALSE
12		latin, qui ex patre filioque procedit. mais l'église grecque retourna encore à son opinion, et sembla la quitter encore dans la réunion	empereur Basile	Rome	FALSE
13		passagère qui se fit avec Eugène IV. Que les hommes apprennent de là à se tolérer les uns les autres. Voilà des variations et des	empereur Basile	Constantinople	FALSE
14		disputes sur un point fondamental, qui n'ont ni excité de troubles, ni rempli les prisons, ni allumé les bûchers. On a blâmé les	Bogoris	Rome	FALSE
15		déférences du pape Jean VIII pour le patriarche Photius ; on n'a pas assez songé que ce pontife avait alors besoin de l'empereur	Bogoris	Constantinople	FALSE
16		Basile. Un roi de Bulgarie, nommé Bogoris, gagné par l'habileté de sa femme qui était chrétienne, s'était converti, à l'exemple de	Clovis	Rome	FALSE
17		Clovis et du roi Egbert.	roi Egbert	Rome	FALSE



What this talk is — and isn't

Is

- The data work behind network analysis in CH/DH
- Two case studies from my own and collaborative research
- A methodological frame, not a methods survey

Isn't

- A graph theory tutorial — no prior knowledge required
- An argument against networks

Roadmap

- Landscape · Argument · Two case studies · Visualization · Q&A



· Prologue ·

The Making of Networks



Network Analysis in Digital/Computational Humanities

A few areas where network analysis has reshaped what we can ask:

- Bibliographic & citation networks — lineages, schools, peripheries
- Social networks — characters in fiction, historical correspondents, archival co-presence
- Archaeological networks — trade, transportation, material connections across sites
- etc.



What they have in common

Across all of these:

- Relations between entities are made explicit
- Every relation requires a decision — what counts as an entity, what counts as a connection
- Those decisions happen before analysis, and they shape what analysis can find

This is where the rest of the talk lives.



Two claims

1. Quality of analysis = quality of data

How insightful your network analysis is → function of the quality of data you build it from.

2. Networks are objects we construct

Data vs. capta (cfr. Drucker 2014 etc.)

Networks are not *found* in the world. Every node, every edge, every weight is the result of a choice — accountable, documentable, and revisable.

Corollary: document your choices.

From text to graph: where do edges come from?

How would you build a graph from this? What's an edge?

	C	E	F	G	H
	is_annotated	text	person	location	at Matteo
1					
2		Le pape Jean VIII le reçut à sa communion, le reconnut, lui écrivit ; et malgré ce huitième concile oecuménique, qui avait anathématisé ce patriarche, le pape envoya ses légats à un autre concile à Constantinople, dans lequel Photius fut reconnu innocent par quatre - cent évêques, dont trois - cent l'avaient auparavant condamné. Les légats de ce même siège de Rome, qui l'avaient anathématisé, servirent eux - mêmes à casser le huitième concile oecuménique. Combien tout change chez les hommes ! Combien ce qui était faux, devient vrai selon les tems ! Les légats de Jean VIII s'écrivent en plein concile ; si quelqu'un ne reconnaît pas Photius, que son partage soit avec Judas. le concile s'écrit, longues années au patriarche Photius, et au patriarche Jean. enfin à la suite des actes du concile on voit une lettre du pape à ce savant patriarche, dans laquelle il lui dit ; nous pensons comme vous... etc. Il est donc clair que l'église romaine et la grecque pensaient alors différemment de ce qu'on pense aujourd'hui. Il arriva depuis que Rome adopta la procession du père et du fils ; et il arriva même qu'en 1274 l'empereur des grecs, Michel Paléologue, implorant contre les turcs une nouvelle croisade, envoya au second concile de Lyon, son patriarche et son chancelier, qui chantèrent avec le concile en latin, qui ex patre filioque procedit. mais l'église grecque retourna encore à son opinion, et sembla la quitter encore dans la réunion passagère qui se fit avec Eugène IV. Que les hommes apprennent de là à se tolérer les uns les autres. Voilà des variations et des disputes sur un point fondamental, qui n'ont ni excité de troubles, ni rempli les prisons, ni allumé les bûchers. On a blâmé les déférences du pape Jean VIII pour le patriarche Photius ; on n'a pas assez songé que ce pontife avait alors besoin de l'empereur Basile. Un roi de Bulgarie, nommé Bogoris, gagné par l'habileté de sa femme qui était chrétienne, s'était converti, à l'exemple de Clovis et du roi Egbert.	Judas	Rome	FALSE
3			Judas	Constantinople	FALSE
4			patriarche Photius	Constantinople	TRUE
5			patriarche Photius	Rome	FALSE
6			patriarche Jean	Constantinople	FALSE
7			patriarche Jean	Rome	FALSE
8			Michel Paléologue	Rome	FALSE
9			Michel Paléologue	Constantinople	PROBABLE
10			Eugène IV	Rome	FALSE
11			Eugène IV	Constantinople	FALSE
12			empereur Basile	Rome	FALSE
13			empereur Basile	Constantinople	FALSE
14			Bogoris	Rome	FALSE
15			Bogoris	Constantinople	FALSE
16			Clovis	Rome	FALSE
17			roi Egbert	Rome	FALSE



The hidden decisions

What is an entity?

- Classes, granularity, disambiguation

What is the context?

- Sentence · paragraph · page · arbitrary chunk

What is an edge?

- Co-occurrence · syntactic · semantic · typed relation

What kind of edge?

- Weighted · directed · labeled

What gets documented?

- Provenance, alternatives considered, version



Three situations researchers find themselves in

A • Graph-ready

- Data is already a graph (e.g., a knowledge graph)
- Work needed: reshape for analysis. Cost lives upstream (ontologies) and in access (SPARQL)

B • Data, no graph

- Data exists; relations not made explicit
- Work needed: enrichment. The most common case; many small consequential choices

C • Not yet data

- Object of study isn't datafied
- Work needed: datafication, then enrichment. Highest cost — you are crafting the data



▪ Case Study 1 ▪

Citation Networks in Classics

Mining Classics Citations from JSTOR

Mining of canonical references from Classics articles in JSTOR, to study scholarly reception of classical texts via citations as a proxy for scholar's attention.

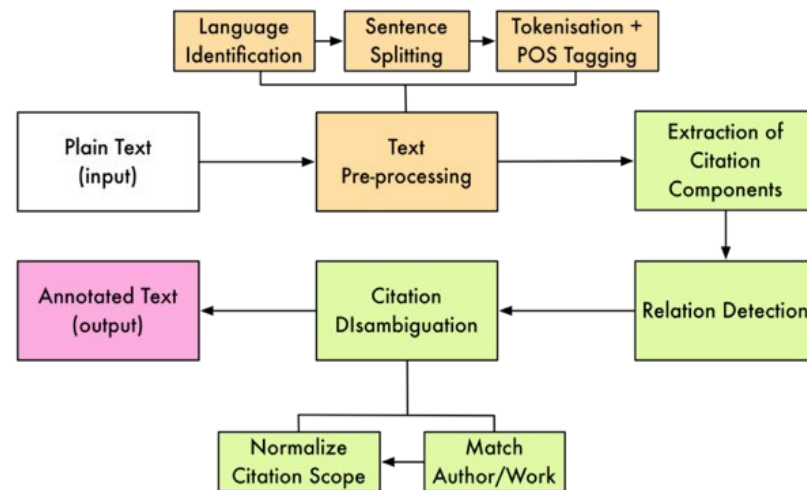
Pre-print: <https://doi.org/10.5281/zenodo.3736455>



Data processing: mining canonical citations

Example of canonical references

first 'block' of arrival scenes in the Telemachy (*Odyssey* 1–4): the arrival of Athena-Mentes at Ithaca (Hom. Od. 1.103–324):⁹ the goddess appears on the threshold of the palace (1.103–4) where she finds the suitors engaged in their respective activities (1.106–12). She is then seen by Telemachus (1.113, 1.118), who rises to accommodate the disguised goddess (1.119–20), takes her by the hand (1.121), and makes her enter (1.125); he welcomes her (1.122–4), takes her spear (1.121 and 1.127–9), and invites her to sit down (1.130–2); the dinner is prepared (1.136–43), consumed (1.149), and concluded (1.150); the visitor's identity is finally revealed (1.169–93) and information is exchanged (1.194–305) before Athena escapes Telemachus' attempt to retain her (1.309–19). This 'classic' hospitality scene highlights Telemachus'



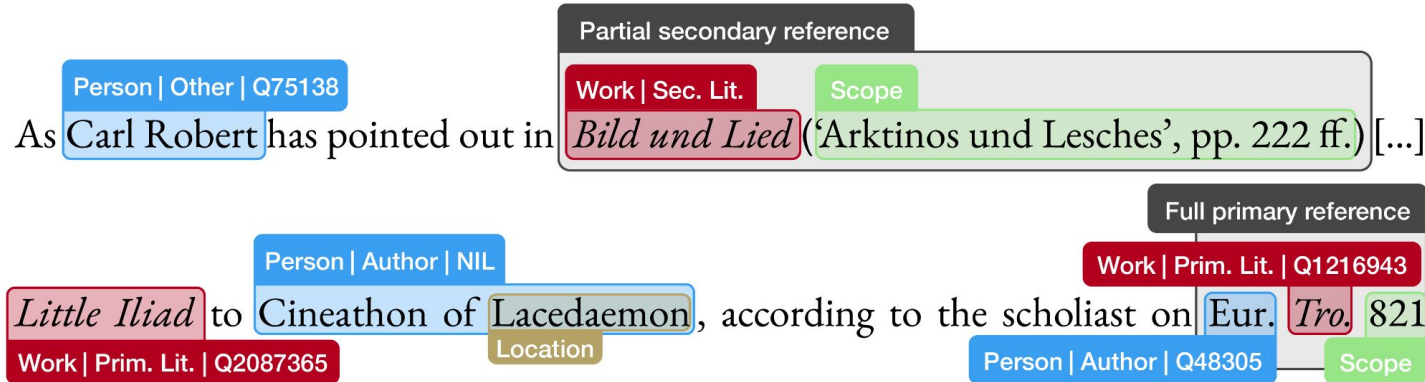
<https://citedloci.org/>

Bibliographic entities

- Primary references
- Secondary references

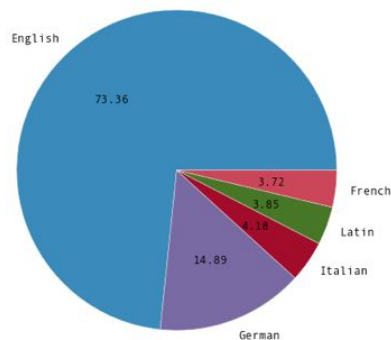
Named entities

- Person
- Place
- Date
- Object
- Work
- Scope



M. Romanello & S. Najem-Meyer, "A Named Entity-Annotated Corpus of 19th Century Classical Commentaries"
<https://doi.org/10.5334/johd.150>

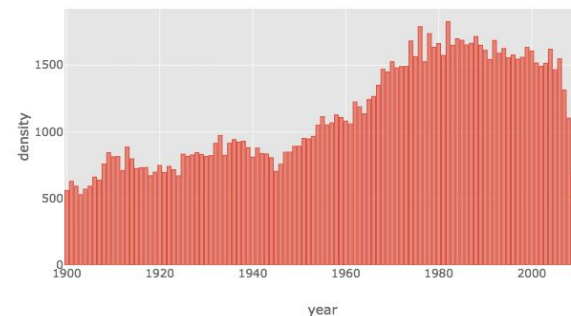
Data: Classics articles in JSTOR



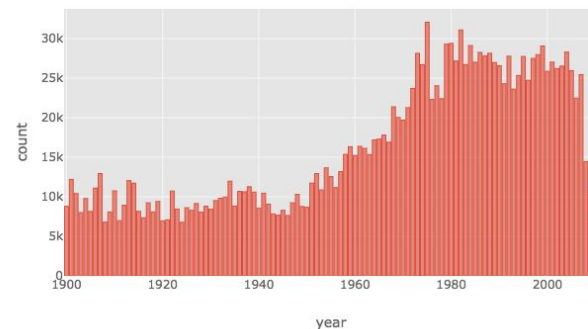
	Number
Total articles	138,821
Successfully processed articles	119,723
Sentences	34,853,399
Tokens	865,075,857
Extracted canonical references	1,649,868
Extracted author mentions	1,448,163
Extracted work mentions	4,665

Table 2: Basic statistics about the JSTOR data.

Number of articles per year in JSTOR (1900-2009)

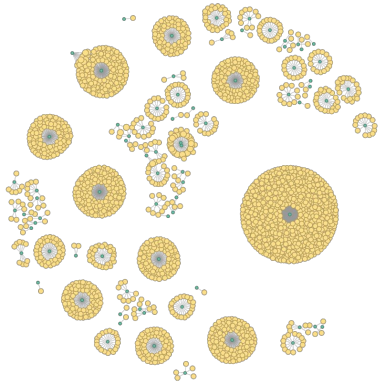


Number of canonical references per year

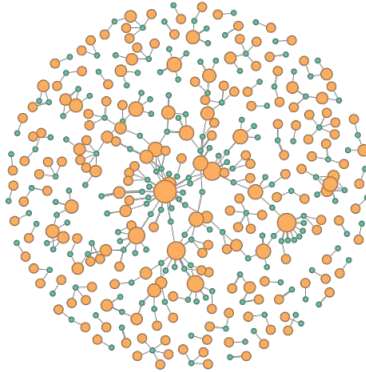


A three-layer citation network

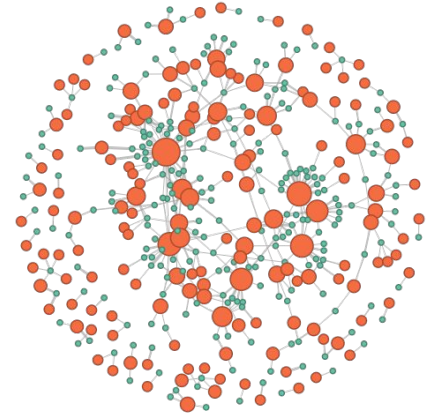
M. Romanello, “Exploring Citation Networks to Study Intertextuality in Classics” in *DHQ* (2016) [[link](#)]



micro-level
publication -> passage

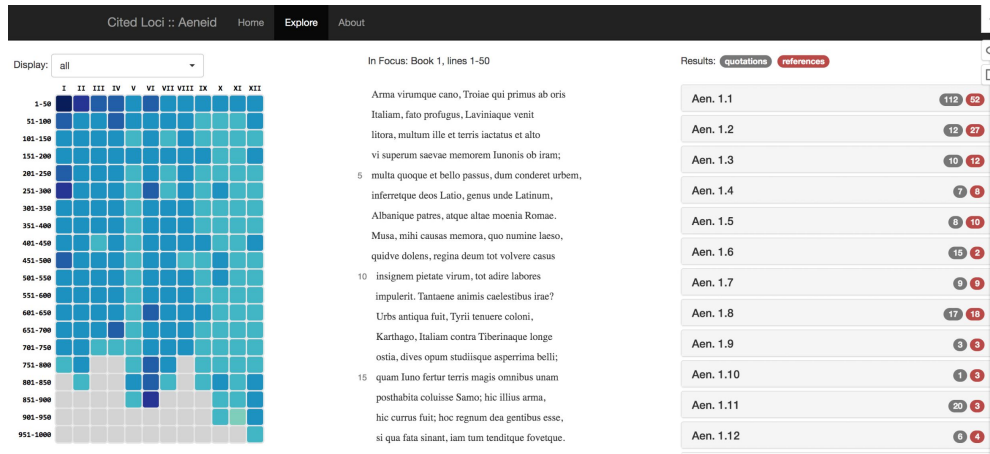


Meso-level
publication -> work



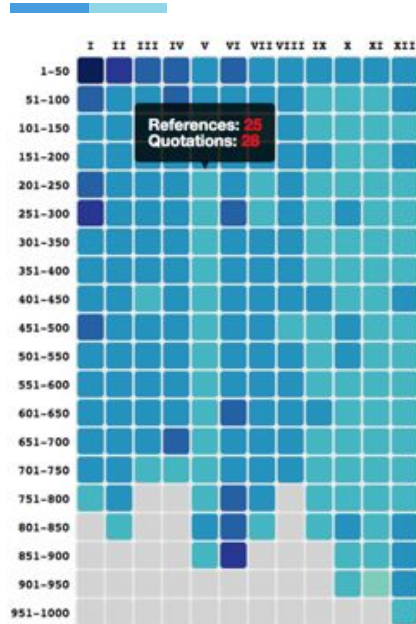
Macro-level
publication -> author

Interactive data visualization: Aeneid in JSTOR



https://mromanello.github.io/Aeneid_in_JSTOR/

<http://labs.jstor.org/blog/cited-loci-of-the-aeneid/>



Distant reading

In Focus: Book 6, lines 251-300

adventante dea. "Procul O procul este, profani,"
conclamat vates, "totoque absistite luco;
260 tuque invade viam, vaginaque eripe ferrum:
nunc animis opus, Aenea, nunc pectore firmo."
Tantum effata, furens antro se immisit aperto;
ille ducem haud timidus vadentem passibus aequat.
Di, quibus imperium est animarum, umbraeque silentes,

Close reading
(primary literature)

Results: quotations references

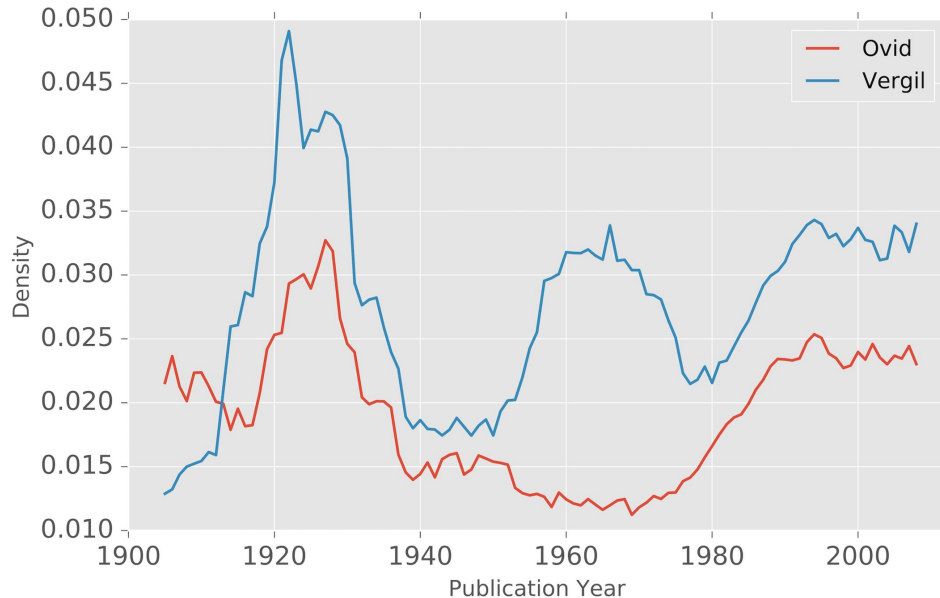
Aen. 6.258	10	7
Aen. 6.260	2	1
Aen. 6.263	1	1
<u>Aen. 6.264</u>	8	13

Proserpina's Tapestry in Claudian's "De raptu": Tradition and Design
MICHAEL VON ALBRECHT
Illinois Classical Studies 1989
 DOI: [10.2307/23064383](https://doi.org/10.2307/23064383)

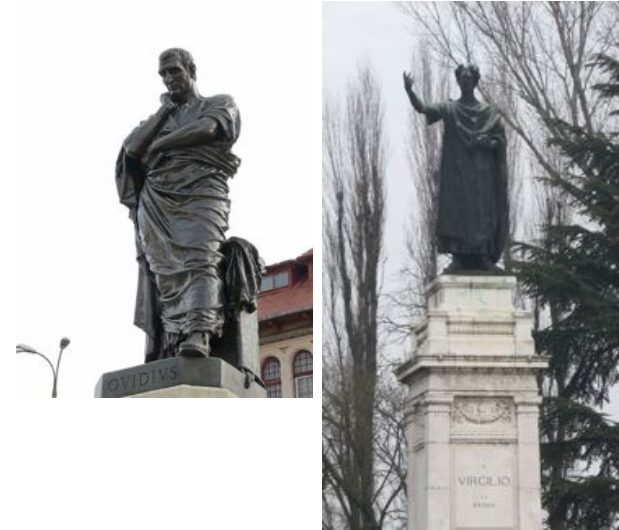
Besides the role of the Sibyl, Claudian also takes the role of Virgil, invoking the gods of the underworld, as his great predecessor had done (1. 20 ; Aen. 6. 264).
 — (reference: 571a493ab634c1c699eed3fa)

Close reading
(secondary literature)

Longitudinal analysis: Waves of reception in Ovid and Vergil



T. Ziolkowski 2009, *Ovid in the Twentieth Century*.



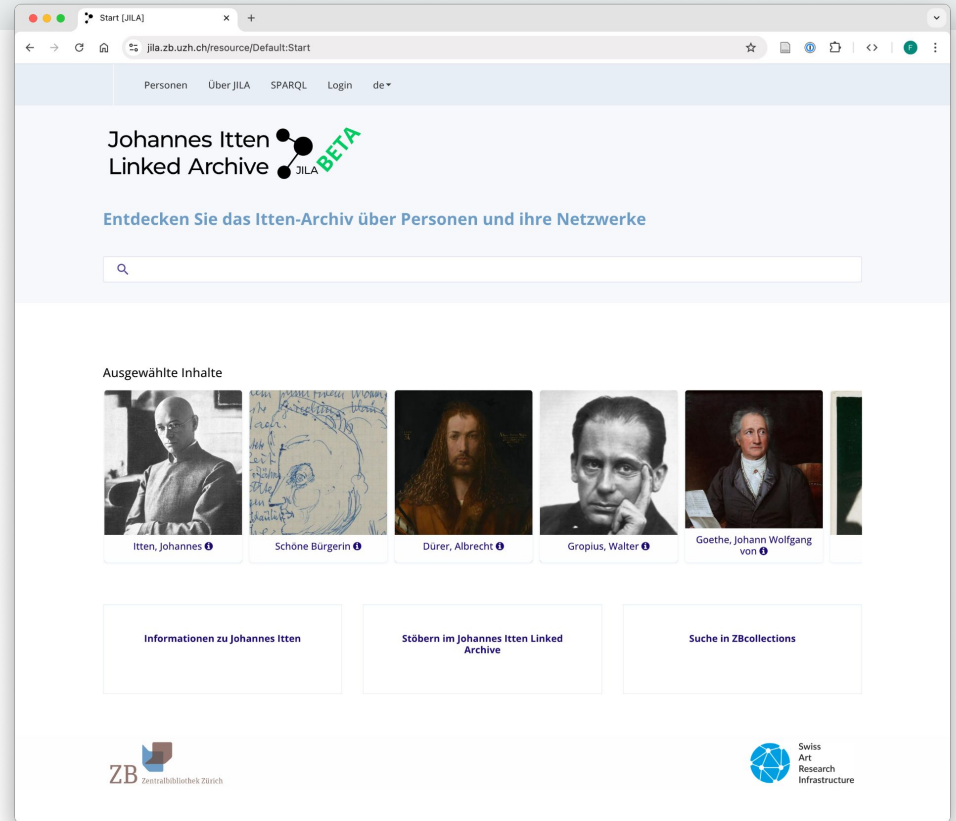


▪ **Case Study 2** ▪

From knowledge graph to network

What is JILA?

Research platform developed in collaboration with the Zurich Central Library
Enables access and exploration of the archive in a Linked Data knowledge graph through a social network centric user interface



[Slide credits: Florian Kräutli (2024)]

Focus: Persons

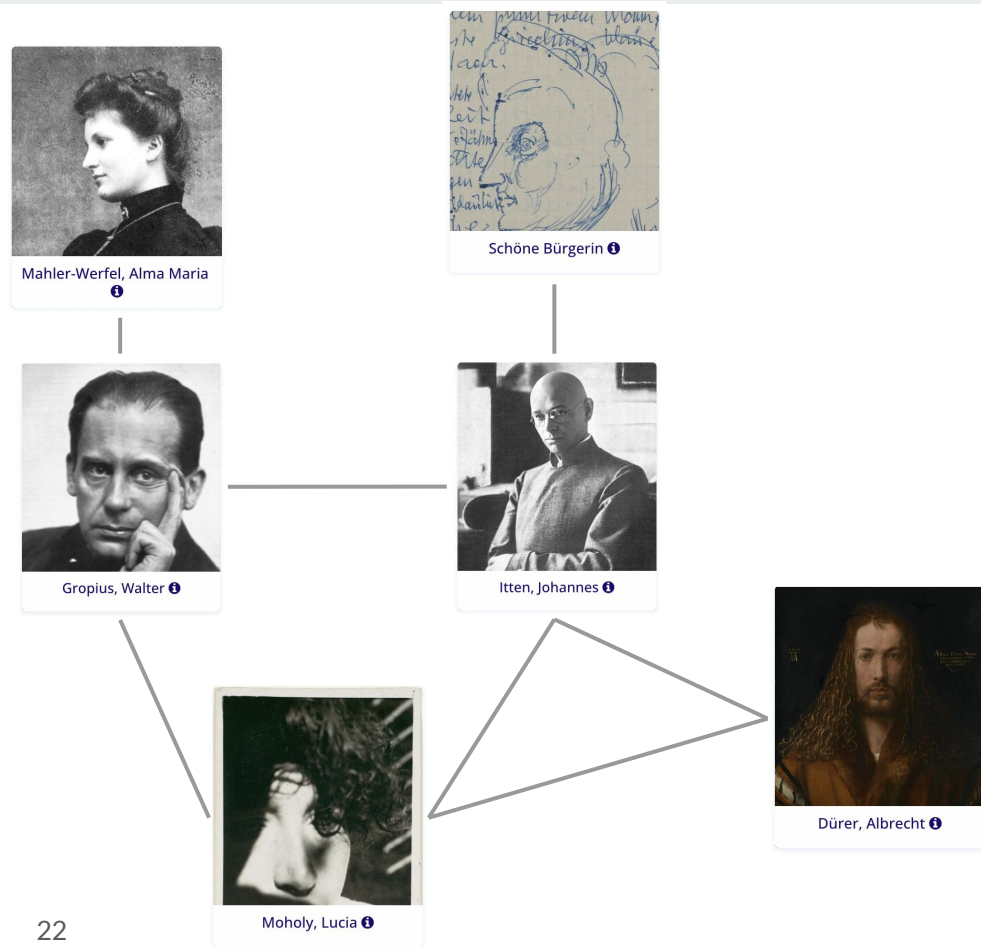
Explore archive based on individuals appearing in material

Identify social network around Itten

Discover material related to individuals and groups of people

Visualise Itten's social network

[Slide credits: Florian Kräutli (2024)]



Core Issue

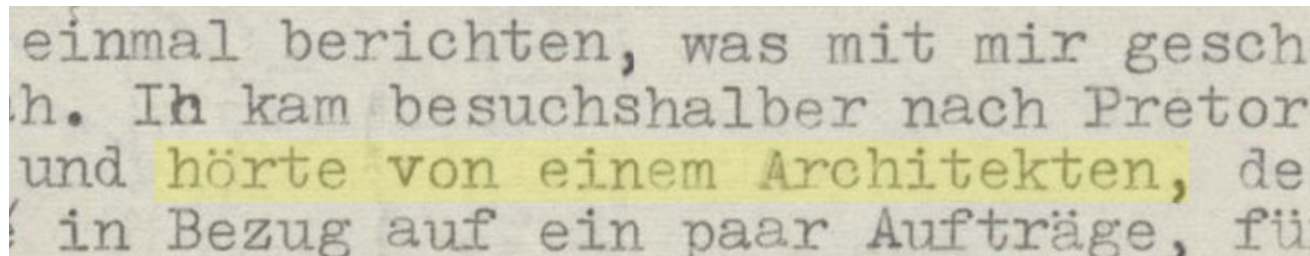
Not all persons can be unambiguously identified

CMI does not support capturing unidentified entities

Individuals mentioned by association or first names only are not captured

Inherent gender bias as men are less affected than women*

[Slide credits: Florian Kräutli (2024)]



einmal berichten, was mit mir gesch
h. Ich kam besuchshalber nach Pretor
und hörte von einem Architekten, de
in Bezug auf ein paar Aufträge, fü

Stauch, Hellmut: Hellmut Stauch an Johannes Itten. 2024. In JILA: Johannes Itten Linked Archive 2024. Zürich: Zentralbibliothek Zürich. Available at: <https://resource.jila.zb.uzh.ch/collection/jila-da5d912d18d146c0af709fe8da54450e>. Accessed 02.02.2024.



Tochter von Adolf Hölzel



Natascha ⓘ



Oskar Schlemmers
Schwiegermutter



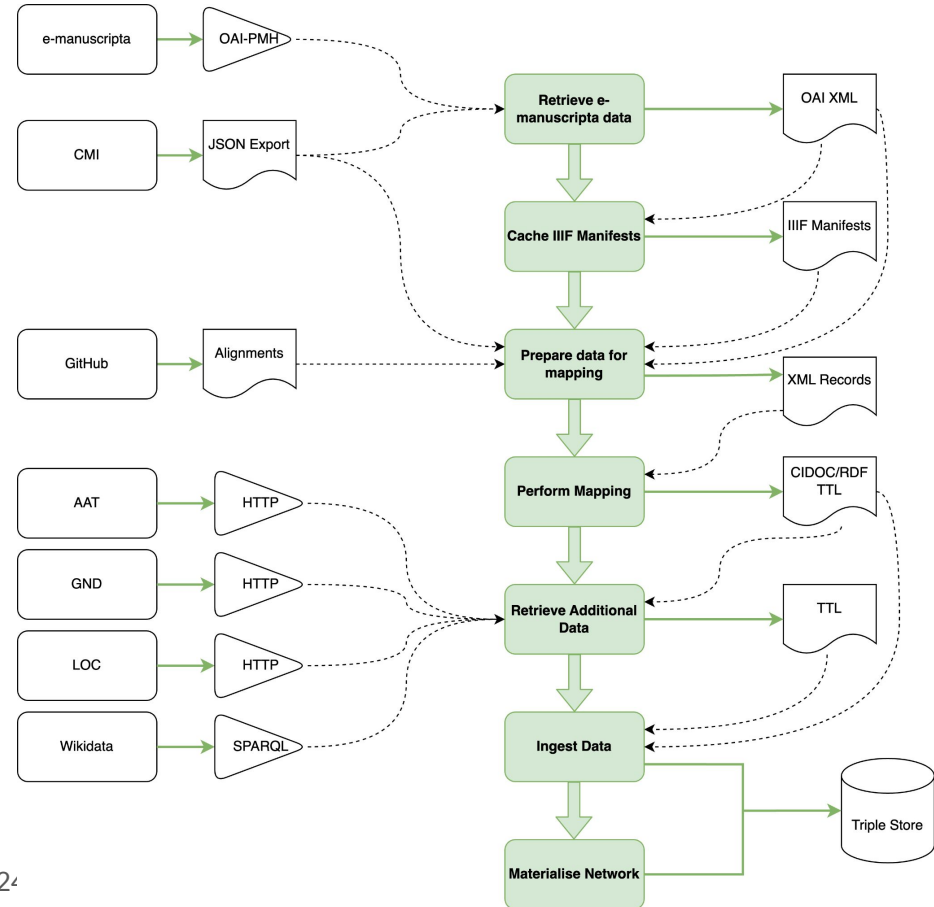
Resi ⓘ

Data Pipeline

Development of a data pipeline that:

- Aggregates several sources
- Converts data to CIDOC/RDF
- Extracts network relationships

[Slide credits: Florian Kräutli (2024)]

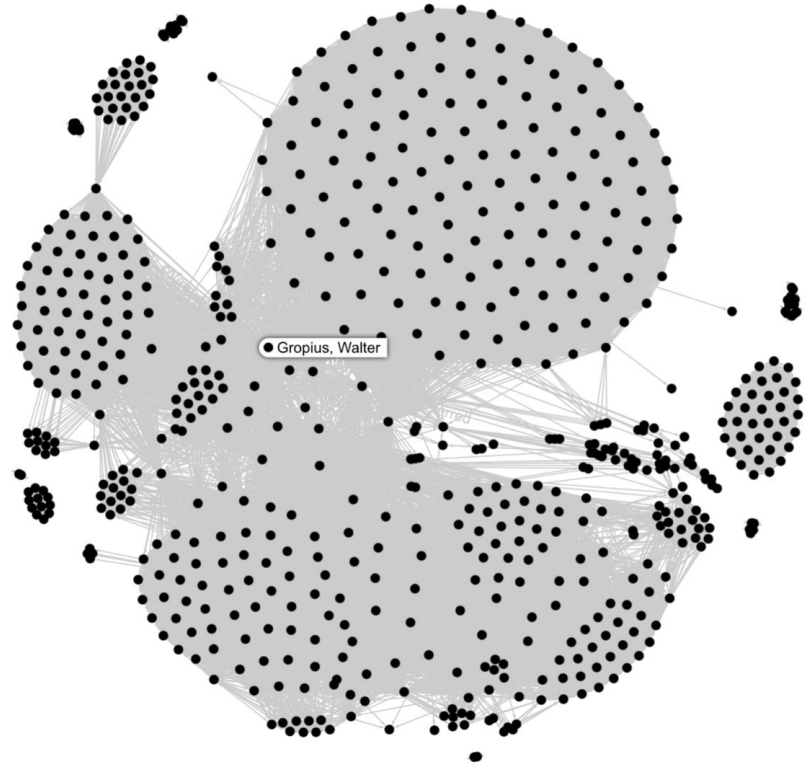


Challenges

Balancing overrepresentation of
dominant figures

Scalability

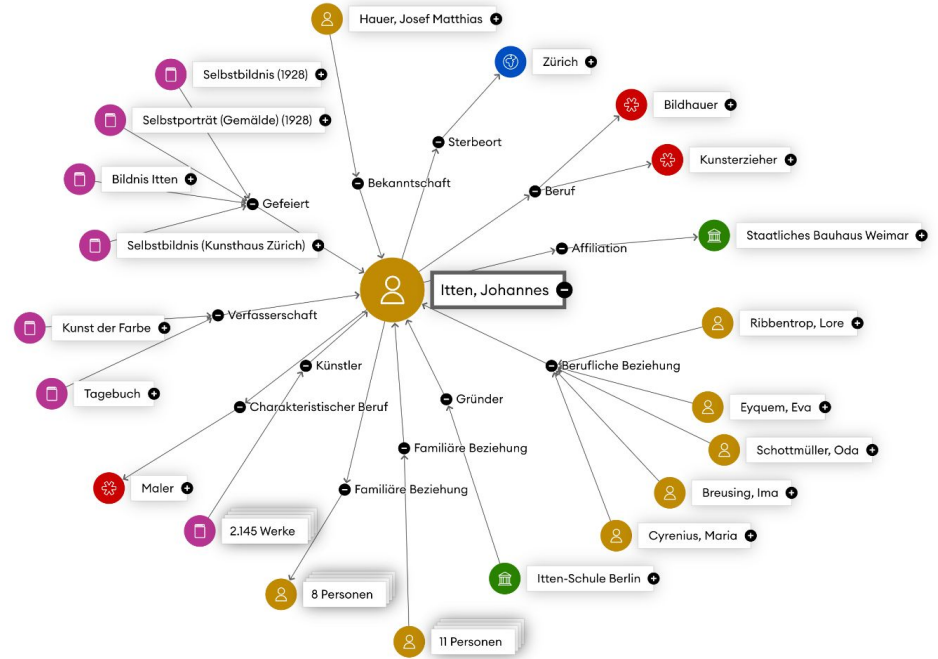
Usability & Interpretability



[Slide credits: Florian Kräutli (2024)]

Precedent

A good example: GND Explorer maintains readability by grouping entities and links
However: like many interactive network visualisation does not scale well



<https://explore.gnd.network/gnd/118710990/relations?term=Johannes+Itten&rows=25&pos=1>

[Slide credits: Florian Kräutli (2024)]

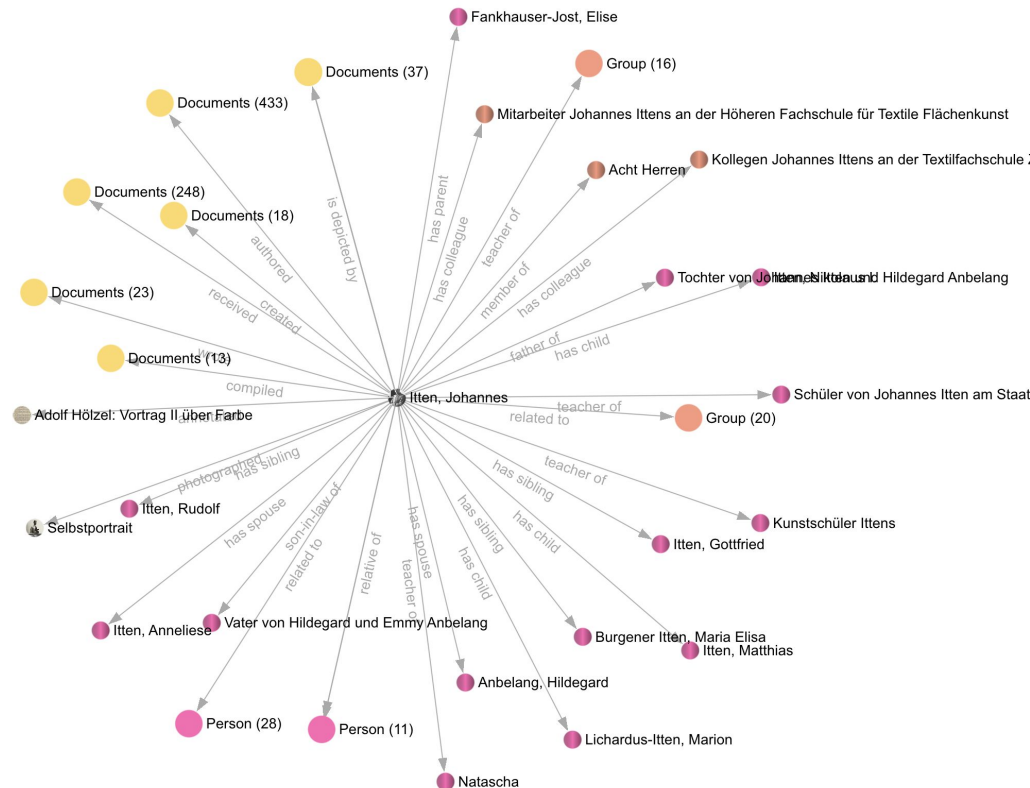
Solution:

GPU based network

visualisation with grouping

mechanism

Multiple modes of navigation



JILA Credits



Swiss Art Research Infrastructure (SARI)

Project Lead (Th. Hänsli, dipl. ETHZ)

Technical Lead (Dr. F. Kräutli)

Software development (Olya Nicolaeva)

Data Engineering (André Ghattas)

Project development (Andrea Notroff)

Project coordination (Stephanie Santschi)

Takin.Solutions (Dr. George Bruseker, Dr. D. Nenova, et al.)

Zellij Semantic Modelling Framework and Application

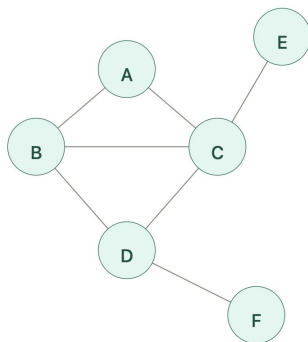


· Epilogue · On (Network) Visualisation

Do you *really* need a network visualisation?

Graph view

Nodes are people, edges are friendships



Matrix view

Filled cell means a tie exists

	A	B	C	D	E	F
A						
B						
C						
D						
E						
F						

Often: no.

- Centrality, community detection, structural metrics – tables and plots beat hairballs
- A viz is for communicating, sometimes for exploring – rarely for analyzing

Both representations encode the exact same information.

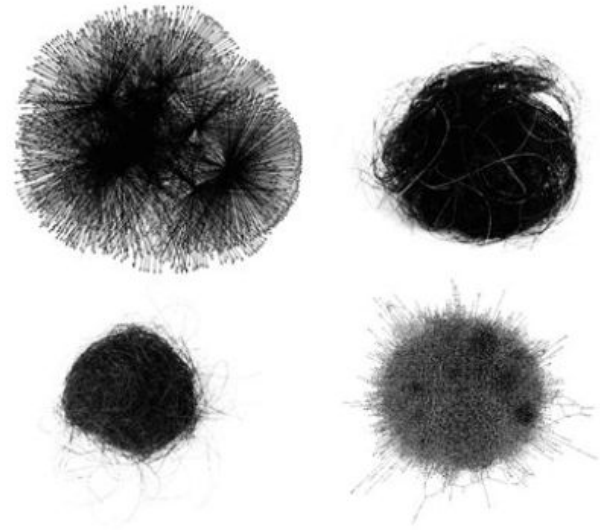
The graph view is intuitive for small networks; the matrix scales better and is the form most algorithms actually operate on.

Networks and hairballs

Graph visualisation is an example of illegible data representation/visualisation.

To overcome this:

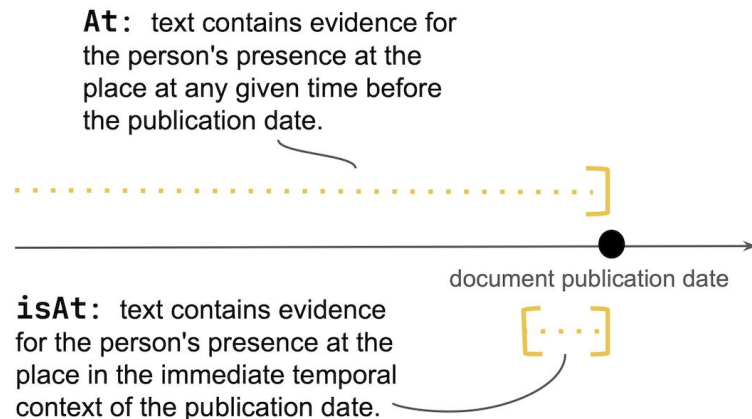
- Use alternative representations (adjacency matrix)
- Filter/remove data to show meaningful patterns
- Focus on graph subsets



Better edges → better networks: HIPE 2026

Where the bottleneck sits today: extracting precise, typed relations from text.

- Most current pipelines stop at named entity recognition + co-occurrence
- Typed relations would let us build networks worth visualizing — and worth analysing
- **HIPE 2026 shared task**: accurate/efficient relation extraction in historical/literary texts



<https://hipe-eval.github.io/HIPE-2026/>



Takeaways

- Networks are constructed — every edge is a decision
- Quality of analysis is downstream of quality of data
- Three situations, three cost profiles: graph-ready • data-but-no-graph • not-yet-data
- Document your choices — they are as much a part of the method as the algorithms
- A visualization is not the goal



Thank you! Questions?

Matteo Romanello (UZH / ETHZ)

romanello@arch.ethz.ch · matteo.romanello@uzh.ch

<http://sari.uzh.ch/>